```
`timescale 1ns/1ps

module tb_async_fifo_8x8;

    reg wr_clk, rd_clk, rst;
    reg wr_en, rd_en;
    reg [7:0] data_in;

    wire [7:0] data_out;
    wire full, empty;
```

```
module async_fifo_8x
    input         wr_clk,
    input         rd_clk,
    input         rst,
    input         wr_en,
    input         rd_en,
    input  [7:0]  data_in,
    output reg [7:0] data_out,
    output reg    full,
    output reg    empty
```

"Handling Multiple Threads in SystemVerilog".

Log | Share

```
TEST 5: Concurrent read/write operations
PASS: Read 80 at time 582000
PASS: Read 81 at time 610000
PASS: Read 82 at time 638000
PASS: Read 83 at time 666000
PASS: Read 84 at time 694000
PASS: Read 85 at time 722000
PASS: Read 86 at time 750000
PASS: Read 87 at time 778000
PASS: Read 88 at time 806000
PASS: Read 89 at time 834000
PASS: Read 8a at time 862000
PASS: Read 8b at time 890000
PASS: Read 8c at time 918000
PASS: Read 8d at time 946000
PASS: Read 8e at time 974000
PASS: Read 8f at time 1002000
PASS: Read 90 at time 1030000
```


```
`timescale 1ns/1ps

module tb_async_fifo_8x8;

    reg wr_clk, rd_clk, rst;
    reg wr_en, rd_en;
    reg [7:0] data_in;

    wire [7:0] data_out;
    wire full, empty;
```

```
module async_fifo_8x
    input         wr_clk,
    input         rd_clk,
    input         rst,
    input         wr_en,
    input         rd_en,
    input  [7:0]  data_in,
    output reg [7:0] data_out,
    output reg    full,
    output reg    empty
```

"Handling Multiple Threads in SystemVerilog".

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```
TEST 1: Write eight words
PASS: FIFO full flag asserted
PASS: FIFO rejects writes while full

TEST 3: Read eight words
PASS: Read 10 at time 274000
PASS: Read 11 at time 302000
PASS: Read 12 at time 330000
PASS: Read 13 at time 358000
PASS: Read 14 at time 386000
PASS: Read 15 at time 414000
PASS: Read 16 at time 442000
PASS: Read 17 at time 470000
PASS: FIFO is empty
PASS: FIFO rejects reads while empty

TEST 5: Concurrent read/write operations
PASS: Read 80 at time 582000
```

```
`timescale 1ns/1ps

module tb_async_fifo_8x8;

    reg wr_clk, rd_clk, rst;
    reg wr_en, rd_en;
    reg [7:0] data_in;

    wire [7:0] data_out;
    wire full, empty;
```

"Handling Multiple Threads in SystemVerilog".

```
module async_fifo_8x8 (
    input        wr_clk,
    input        rd_clk,
    input        rst,
    input        wr_en,
    input        rd_en,
    input  [7:0] data_in,
    output reg [7:0] data_out,
    output reg   full,
    output reg   empty
```

Log | Share

```
PASS: Read 8b at time 890000
PASS: Read 8c at time 918000
PASS: Read 8d at time 946000
PASS: Read 8e at time 974000
PASS: Read 8f at time 1002000
PASS: Read 90 at time 1030000
PASS: Read 91 at time 1058000
PASS: Read 92 at time 1086000
PASS: Read 93 at time 1114000
PASS: FIFO is empty

TEST PASSED: All checks successful
```